INCH-POUND
MIL-DTL-24231/9E(SH)
26 April 2013
SUPERSEDING
MIL-C-24231/9D(SH)
15 May 1987

DETAIL SPECIFICATION SHEET

CONNECTOR, SINGLE CABLE, TYPE III, SEVEN- AND NINE-CONDUCTOR

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24231.

REQUIREMENTS:

1. Dimensions and configuration: see figures 1 through 8.

2. The spring lockwasher, part number M24231/9-023, shall be in accordance with NASM35338-126 except that the inside diameter (id) shall be 0.796.

3. Refer to MIL-DTL-24231/5 for alternate method of manufacturing connector body and installing guide pin M24231/5-071.

4. For welding of the pin, item 23, see MIL-DTL-24231/5.

5. Unless otherwise specified (see figure 8), all dimensions are in inches.

6. Tolerances shall be as follows, or as stated:

a. Decimals – ±0.005 inch.

b. Fractions – ±1/64 inch.

c. Angular dimensions – ±½ degree.

d. Tolerances for rubber packing should follow the guidance provided in table XLI (RMA Drawing Designation A2 Precision) of MIL-HDBK-149.

7. The flange outside diameter is intended to coincide with commonly available 2¾ inch bar stock. The dimension may vary after machine clean-up.

8. The gland nut shall be tightened until the packing tightens against the wires and cables. This can be determined by moving the wires and cables gently back and forth until the wire and cable is seized by the packing. The gland nut shall be turned approximately one-fourth of a turn more to hold wires and cables in place. Do not overtighten the gland nut or attempt to tighten down metal to metal, as this can damage the wires and cables.

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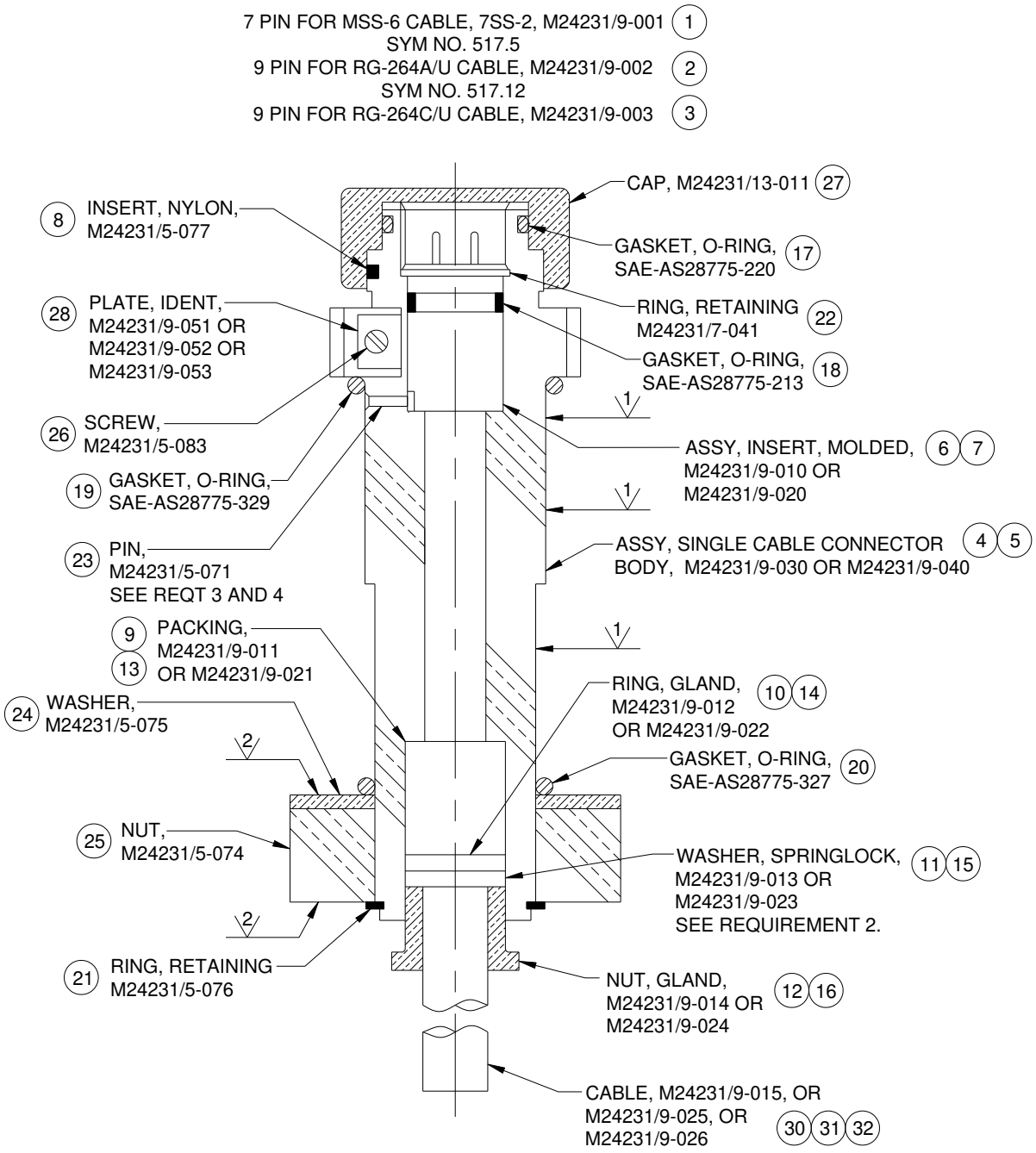
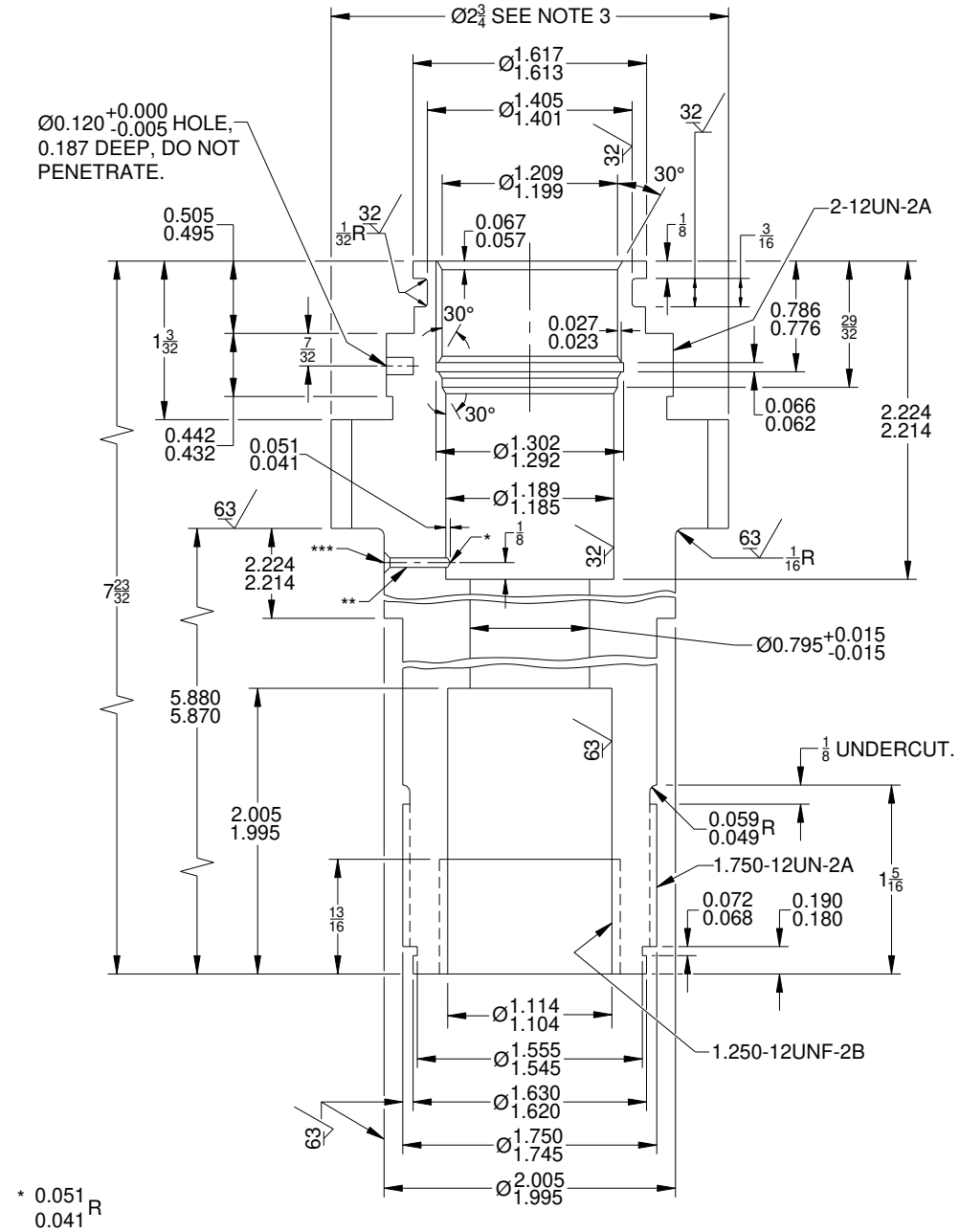


FIGURE 1. Connector, single cable, seven- and nine-conductor.

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* 0.051 0.041 R

** PIN, Ø 0.097 0.087 LG. TO SUIT M24231/5-071

*** WELD.

125 ALL OVER EXCEPT AS SHOWN.

FIGURE 2. Assembly, single cable connector body – M24231/9-030, M24231/9-040. (4)(5)

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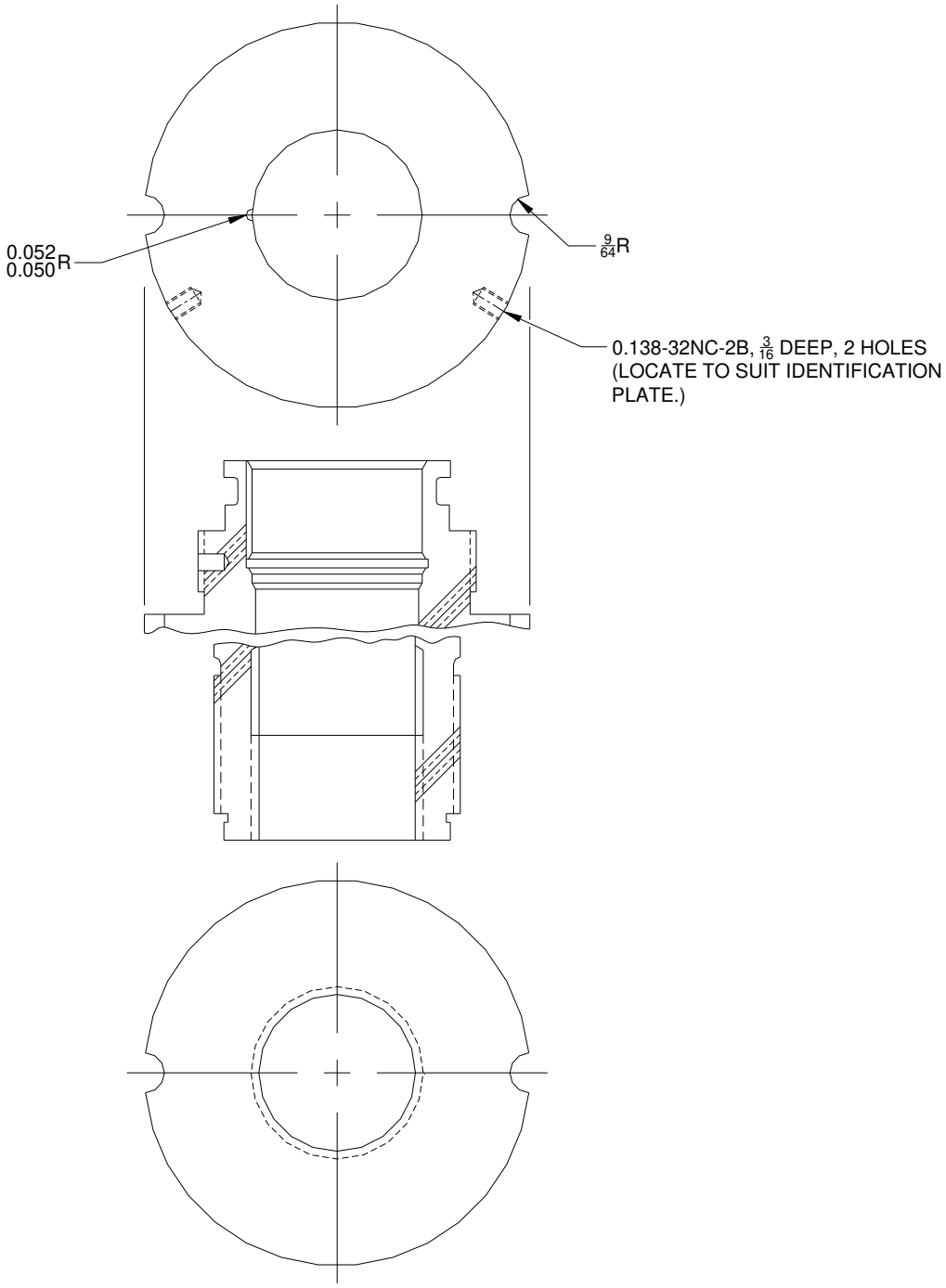


FIGURE 2. Assembly, single cable connector body – M24231/9-030, M24231/9-040 – Continued. (4)(5)

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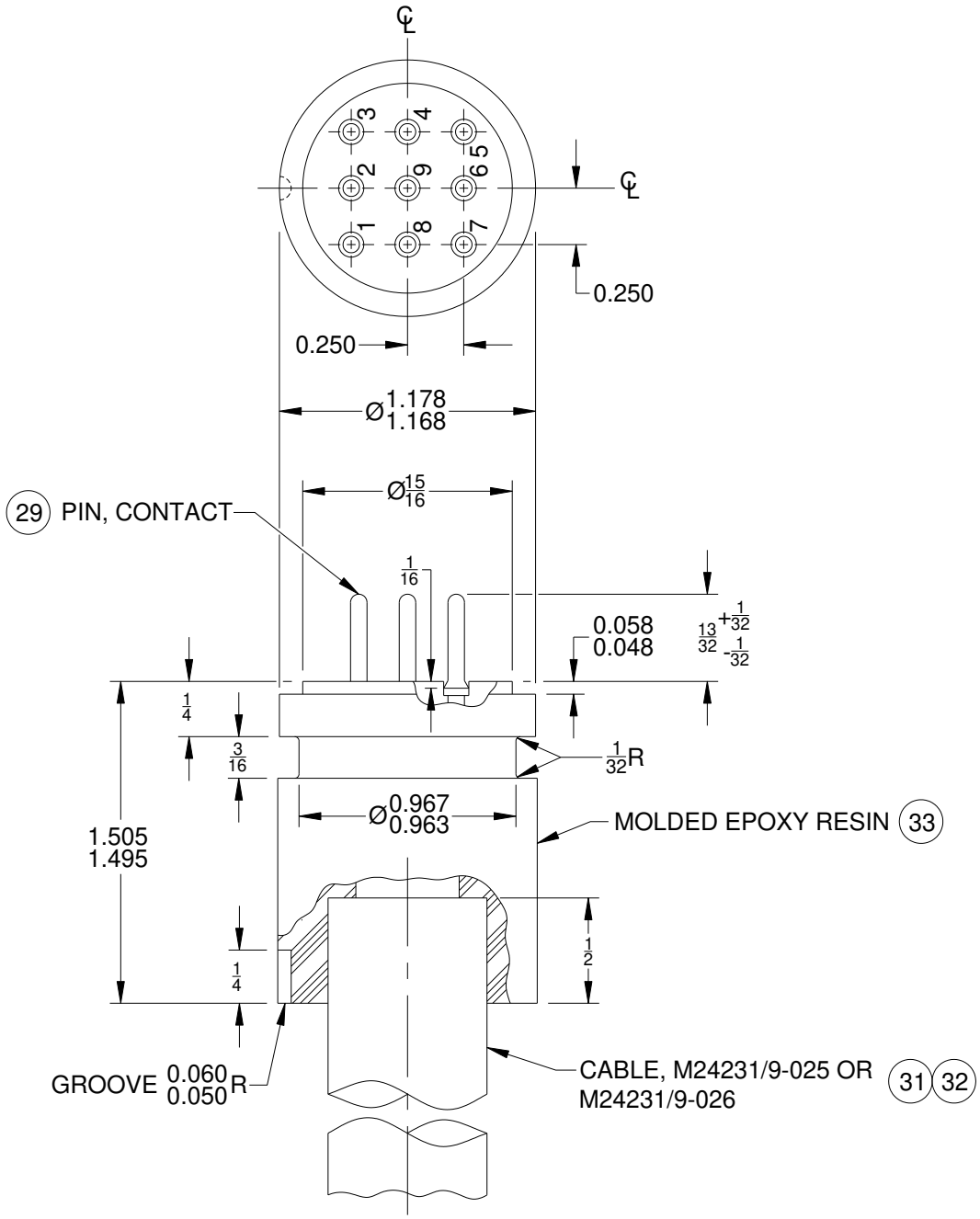


FIGURE 3. Assembly, insert, molded – M24231/9-020. (7)

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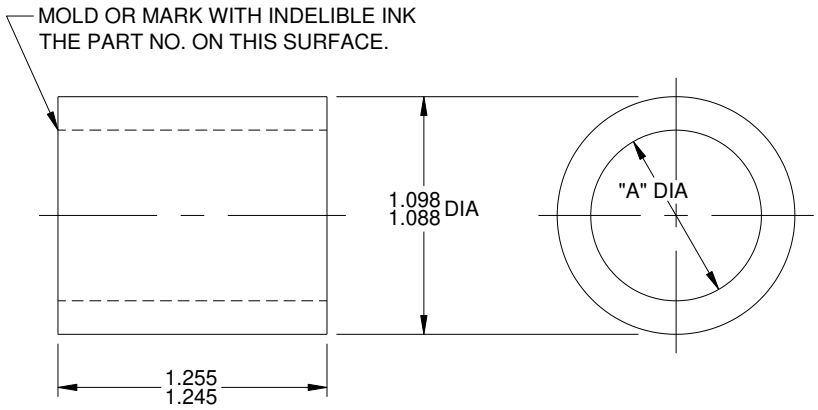
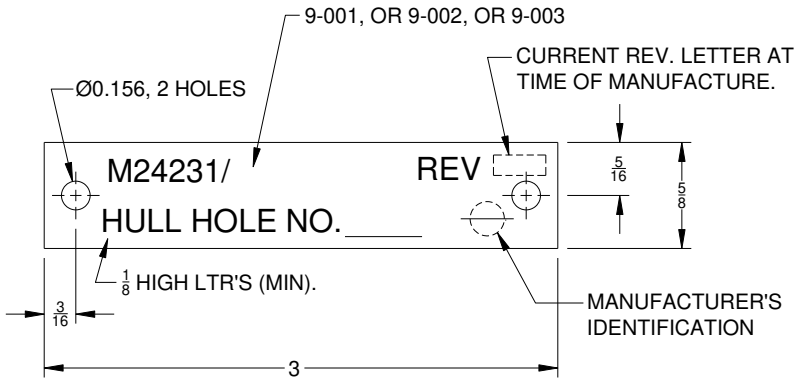


FIGURE 4. Packing – M24231/9-011 (A=0.600), M24231/9-021 (A=0.782). (9)(13)



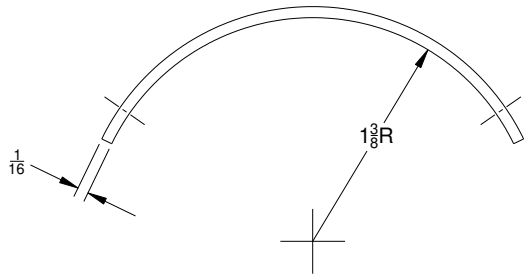


FIGURE 5. Plate, identification - M24231/9-051 (for M24231/9-001), M24231/9-052 (for M24231/9-002), M24231/9-053 (for M24231/9-003). (28)

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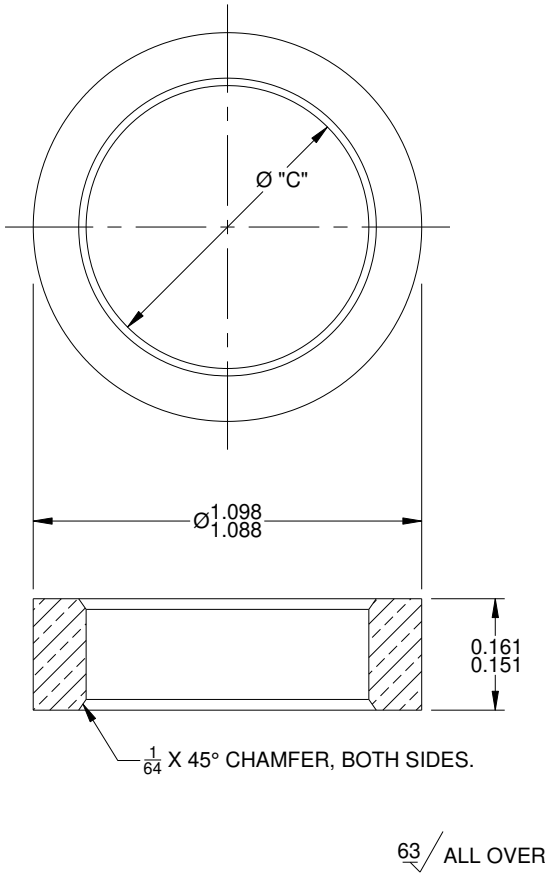


FIGURE 6. Ring, gland – M24231/9-012 (C=0.671), M24231/9-022 (C=0.796). 10 14

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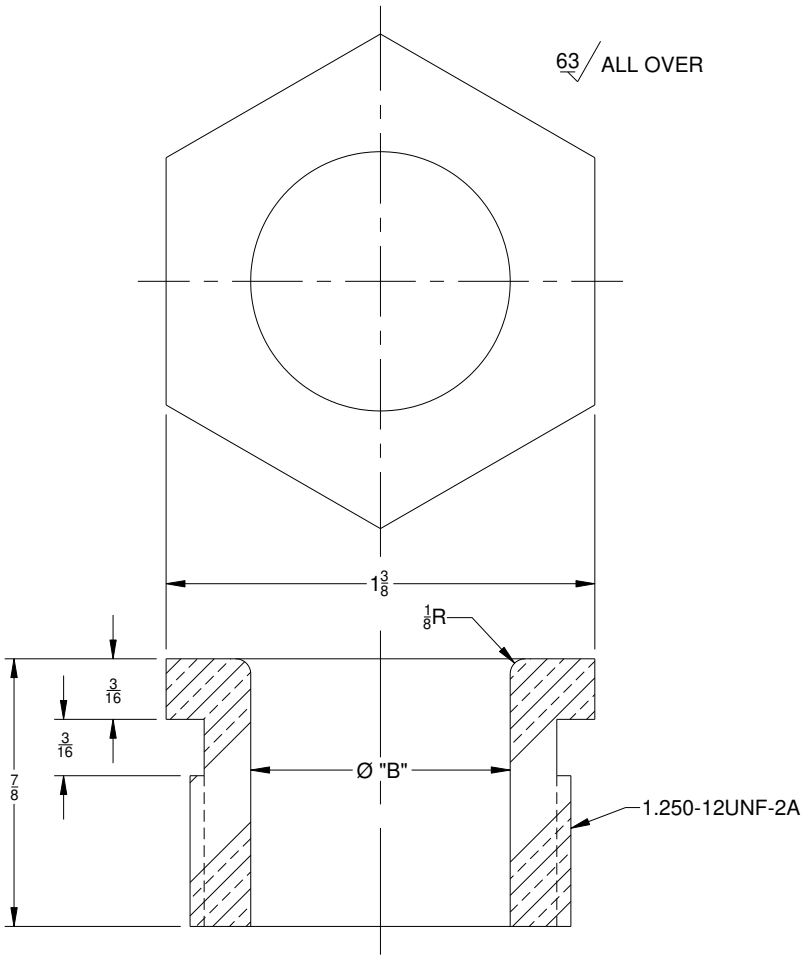


FIGURE 7. Nut, gland – M24231/9-014 (B=0.671), M24231/9-024 (B=0.796). 12 16

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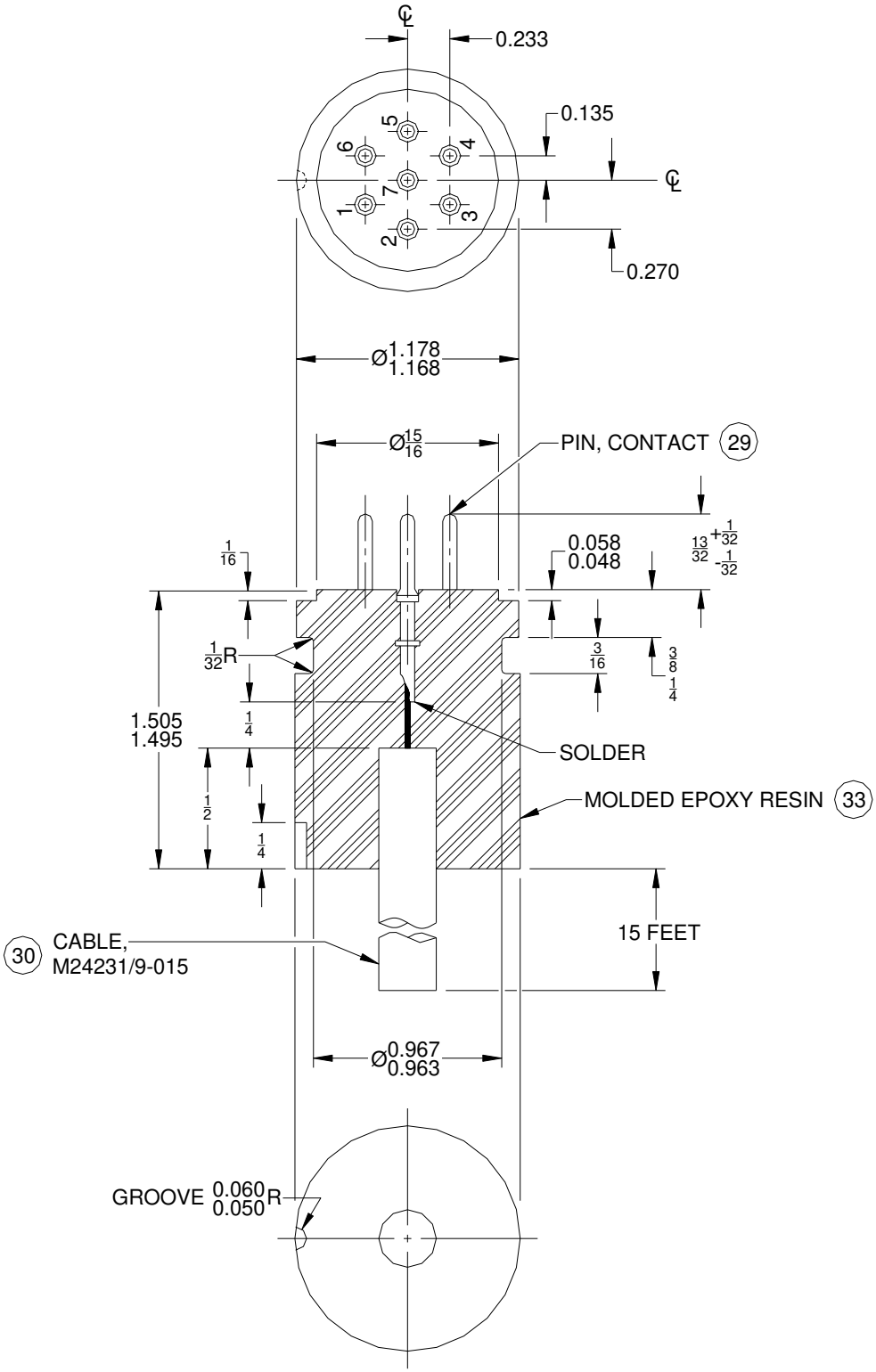


FIGURE 8. Assembly, insert, molded (7-conductor) – M24231/9-010. (6)

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TABLE I. Parts list.

Item no. 1/	Description 2/	Quantity for M24231/9- 3/			Material/specification	Remarks
		001	002	003		
1	Connector assembly, single cable, symbol no. 517.5	1			-----	7-pin for cable MSS-6 or 7SS-2, see figure 1.
2	Connector assembly, single cable, symbol no. 517.12		1		-----	9-pin for cable RG-264 A/U, see figure 1.
3	Connector assembly, single cable			1	-----	9-pin for cable RG-264 C/U, see figure 1.
4	Connector body, single cable	1			Ni-Cu-Al, QQ-N-286	See figure 2 and continuation of figure 2 and requirement 3.
5	Connector body, single cable		1	1	Ni-Cu-Al, QQ-N-286	See figure 2 and continuation of figure 2 and requirement 3.
6	Insert assembly, molded 7-conductor	1			See items 29, 30, and 33.	See figure 8.
7	Insert assembly, molded 9-conductor		1	1	See items 29, 31 or 32, and 33.	See figure 3.
8	Insert	1	1	1	M24231/5-077	See figure 1.
9	Packing, M24231/9-011, Class A, durometer 55±5	1			Rubber, ZZ-R-768	See figure 4.
10	Ring, gland, M24231/9-012	1			Ni-Al-BRZ, ASTM B150/B150M	See figure 6.
11	Washer, lock, M24231/9-013	1			MS35338-126	See figure 1.
12	Nut, gland M24231/9-014	1			Ni-Al-BRZ, ASTM B150/B150M	See figure 7.
13	Packing, Class A, durometer 55±5, M24231/9-021		1	1	Rubber, ZZ-R-768	See figure 4.
14	Ring, gland, M24231/9-022		1	1	Ni-Al-BRZ, ASTM B150/B150M	See figure 6.
15	Washer, spring-lock, M24231/9-023		1	1	MS35338-126	See figure 1 and requirement 2.
16	Nut, gland, M24231/9-024		1	1	Ni-Al-BRZ, ASTM B150/B150M	See figure 7.
17	Gasket, O-ring	1	1	1	SAE-AS28775-220 4/	Lubricate with silicone compound, SAE-AS8660, see figure 1.
18	Gasket, O-ring	1	1	1	SAE-AS28775-213 4/	See item 17.
19	Gasket, O-ring	1	1	1	SAE-AS28775-329 4/	See item 17.
20	Gasket, O-ring	1	1	1	SAE-AS28775-327 4/	See item 17.
21	Ring, retaining	1	1	1	M24231/5-076	See figure 1.

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TABLE I. Parts list – Continued.

Item no. [1/]	Description [2/]	Quantity for M24231/9- [3/]			Material/Specification	Remarks
		001	002	003		
22	Ring, retaining	1	1	1	M24231/7-041	See figure 1.
23	Pin, guide	1	1	1	M24231/5-071	See requirement 3.
24	Washer	1	1	1	M24231/5-075	See figure 1.
25	Nut	1	1	1	M24231/5-074	See figure 1.
26	Screw	1	1	1	M24231/5-083	See figure 1.
27	Cap, pressure-proof	1	1	1	M24231/13-011	See figure 1.
28	Plate, identification	1	1	1	Corrosion-resistant steel or aluminum, MIL-DTL-15024	See figure 5.
29	Pin, contact	7	9	9	MIL-DTL-24231	See figure 8.
30	Cable, 7SS-2	15 ft			MIL-DTL-915/8	See figure 1.
31	Cable, RG-264 A/U	As required			[5/]	See figure 1 and figure 3.
32	Cable, RG-264 C/U	As required			[5/]	See figure 1 and figure 3.
33	Epoxy resin, molded	As required			Cast epoxy resin compound [6/]	See figure 3 and figure 8.

NOTES:

1/ Item numbers are the same as encircled numbers shown on figures.

2/ See applicable figure or table listed in remarks column for part numbers and dimensions.

3/ Quantity is for one complete assembly of each size.

4/ Nitrile (Buna N); Parker Hannifin Corporation (part number N0756-75), O-ring Division, 2360 Palumbo Drive, Lexington, KY 40509, CAGE 02697, or equal.

5/ Formerly shown in MIL-DTL-23020 and MIL-C-23020/4.

6/ Arco Set 200, M.M. Reif and Co., Ivyland, PA, or PR610-TCF (Chockfast Orange), ITW Philadelphia Resins Corp., Montgomeryville, PA, or equal.

CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing activity:
Navy – SH
(Project 5935-2011-058)

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